

# PHANPHUM PRATHUMSUWAN

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## EDUCATION

**Mahidol University, Faculty of Information and Communication Technology (ICT),**  
*Bachelor of Science in Software Engineering*

Aug 2022 – May 2026

## EXPERIENCE

### Full-Stack Developer for DormiSpace: Dormitory Management Platform

Aug 2025 – Apr 2026

- Designed and developed scalable RESTful APIs using Spring Boot for authentication, contract lifecycle management, and a multi-tenant billing system, with secure role-based access control and optimized PostgreSQL schemas.
- Led Flutter frontend development for the mobile application, translating UI/UX requirements into maintainable screens, reusable components, and iterative feature improvements.
- Diagnosed production-level issues involving API responses, data integrity, and system reliability, while maintaining GitHub Actions CI/CD pipelines to support stable release delivery.

### Research Intern at ReaLearn Laboratory — *Tohoku University, Japan*

May 2025 – Aug 2025

- Designed and implemented an automated benchmarking framework to evaluate reliability and failure behavior across seven LLMs, improving evaluation coverage and support for robust AI deployment.
- Analyzed model outputs to identify failure patterns and reliability gaps, producing insights that informed more stable AI system behavior.

### Research Intern at Ishii Laboratory — *Japan Advanced Institute of Science and Technology (JAIST), Japan*

Jun 2024 – Aug 2024

- Applied formal verification techniques using Lustre/Kind2 and Simulink/SLDV to analyze system correctness, safety properties, and complex system behavior.
- Translated and modelled system specifications across platforms, identifying state-space complexity limitations and producing insights to support technical documentation, system planning, and design decisions.

## PROJECTS

### GoodPoliz: AI-Assisted Police Operations System

- Implemented Flutter automated UI tests using the integration\_test framework and Flutter Drive runner on Flutter Web (Chrome), covering happy and unhappy path scenarios across three core workflows
- Wrote backend unit tests (JUnit + Mockito) applying input space partitioning and logic coverage techniques; conducted manual system testing and produced a requirement traceability matrix to ensure test coverage aligned with functional requirements

### Gemini Telescope Control System

- Designed and implemented a web-based control system integrating modern RESTful APIs with a legacy Observatory Control System (OCS), enabling seamless communication between new and existing components
- Modelled system architecture using UML diagrams to support technical design planning and stakeholder

### Online Tutor Search and Management Website

- Designed and implemented backend logic and relational database schemas (SQL) for managing tutor profiles and booking workflows
- Optimized query performance and matching algorithms to improve system efficiency and response time

## ACTIVITY

### Participant in Mahidol AI Hackathon 2 — *Faculty of Information and Communication Technology*

- Collaborated in a cross-functional team to design and implement an AI-based solution for addressing healthcare service challenges
- Managed data preparation and model integration, ensuring smooth interaction between data pipelines and application logic

## SKILLS & LANGUAGES

**Programming:** C, Java, Python, Dart, JavaScript, SQL

**Backend & Database:** Spring Boot, RESTful API Design, PostgreSQL, MySQL, Firebase

**Cloud & Third-Party Services:** AWS, Google Cloud Vision OCR, Omise Payment Gateway

**Testing & DevOps:** JUnit, Mockito, Flutter Unit/Integration Testing, Robot Framework, JMeter, GitHub Actions, Docker

**Systems & Reliability:** Linux, Kibana, application monitoring, incident troubleshooting, root cause analysis

**Languages:** Thai (Native), English (IELTS 7.0, TOEIC 895), Japanese (JLPT N5)